How TimesFM-3 makes a forecast

Google · 331M parameters · decoder-only transformer · zero-shot

1. One row per series

Up to 32 rows: targets, past-only and known-ahead covariates. A ? is a value not known yet.

past *future*

Target temperature

Known ahead day length

Past only NAO index

Five metal prices go in together as five target rows.

2. Scale, then build tokens

Each row is scaled as it goes, with the mean and standard deviation up to each patch. A token holds a patch of 32 steps and the next 64 steps, which only known-ahead rows can fill.

target

known

patch *next 64 steps*

residual MLP

one token: 1,280 numbers

A row with a strong straight-line trend is detrended first.

3. Twenty blocks, two kinds of attention

RMSNorm sits before and after every layer, with a shortcut around it. Future patches go in as masked tokens.

past *future*

temperature

day length

NAO index

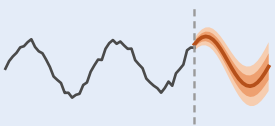
Sequence attention: along the row, this patch and earlier ones (RoPE)

Variate attention: down the column, every row at the same patch

Feed forward: each token on its own, width 1,280, ReLU

4. Every horizon in one pass

A linear head turns each future token of a target row into **64 steps × 9 quantiles**.



median

60% band

80% band

Tokens 32 steps apart overlap by 32 steps; the overlaps are blended. The trend and scaling are undone and the median is the point forecast.

Variate attention lets each target read its covariates and the other targets at every patch.

20 blocks · width 1,280 · patches of 32 in, 64 out · up to 32 rows · context up to 15,360 steps

In the notebooks: last 2,048 days or full monthly history · all covariates and the five metals together · weights for non-commercial use